SmartREC-RAG: Training-free LLM Enhenced/Augmented Recommendation Systems

SURF-2025-0537

SURF Summer Undergraduate Research Fellowship

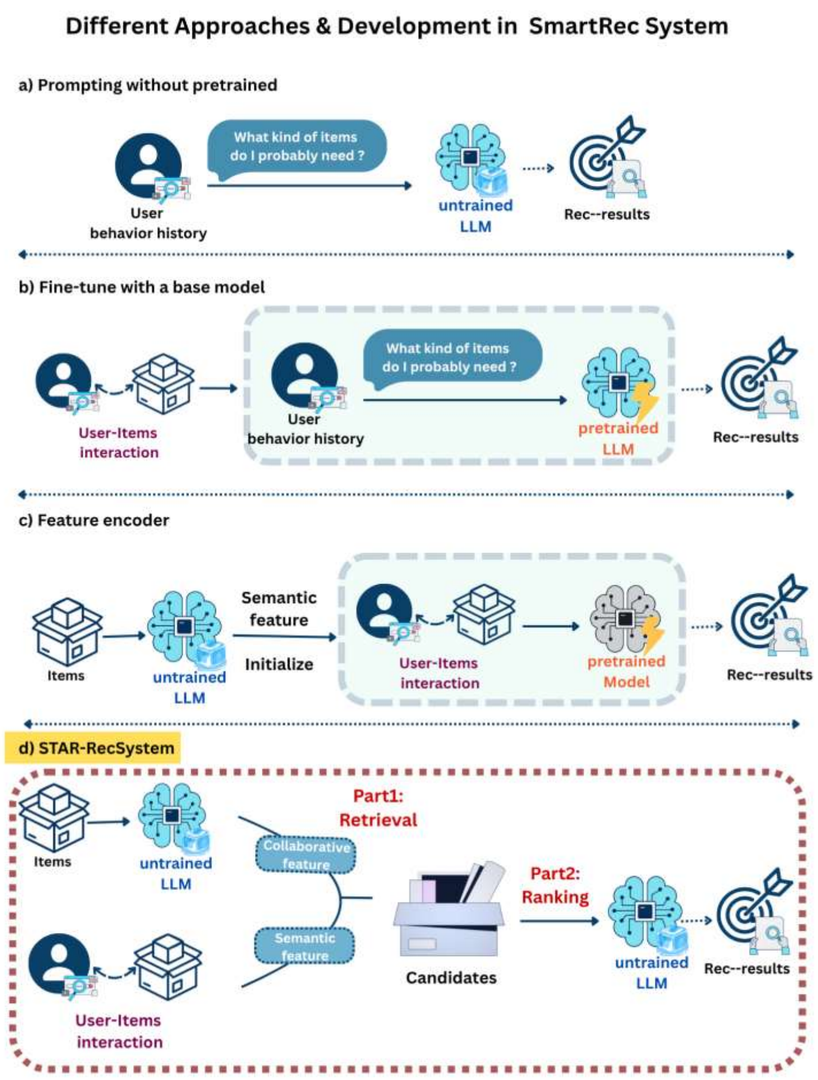
Xuanzheng.Lin, Qian.Peng, Yifan.Wang, Zihan.Zhu, Kaihang.Kong, Zhongyang.Liu

Supervisor: Dr. Bin Chen

Abstract

This research introduces **STAR-RecSystem**, a novel training-free recommendation framework. It leverages the inherent semantic understanding of large language models (LLMs) and collaborative filtering from user behavior to generate high-quality recommendations. The system first constructs semantic and collaborative relationship matrices. It then uses a scoring rule to retrieve an initial set of candidate items. Finally, an LLM ranks these candidates to refine the results. This approach bypasses the need for model training, directly using LLM capabilities for both candidate retrieval and sophisticated ranking. Experiments on datasets like Amazon Reviews show the STAR model significantly outperforms traditional KNN baselines.

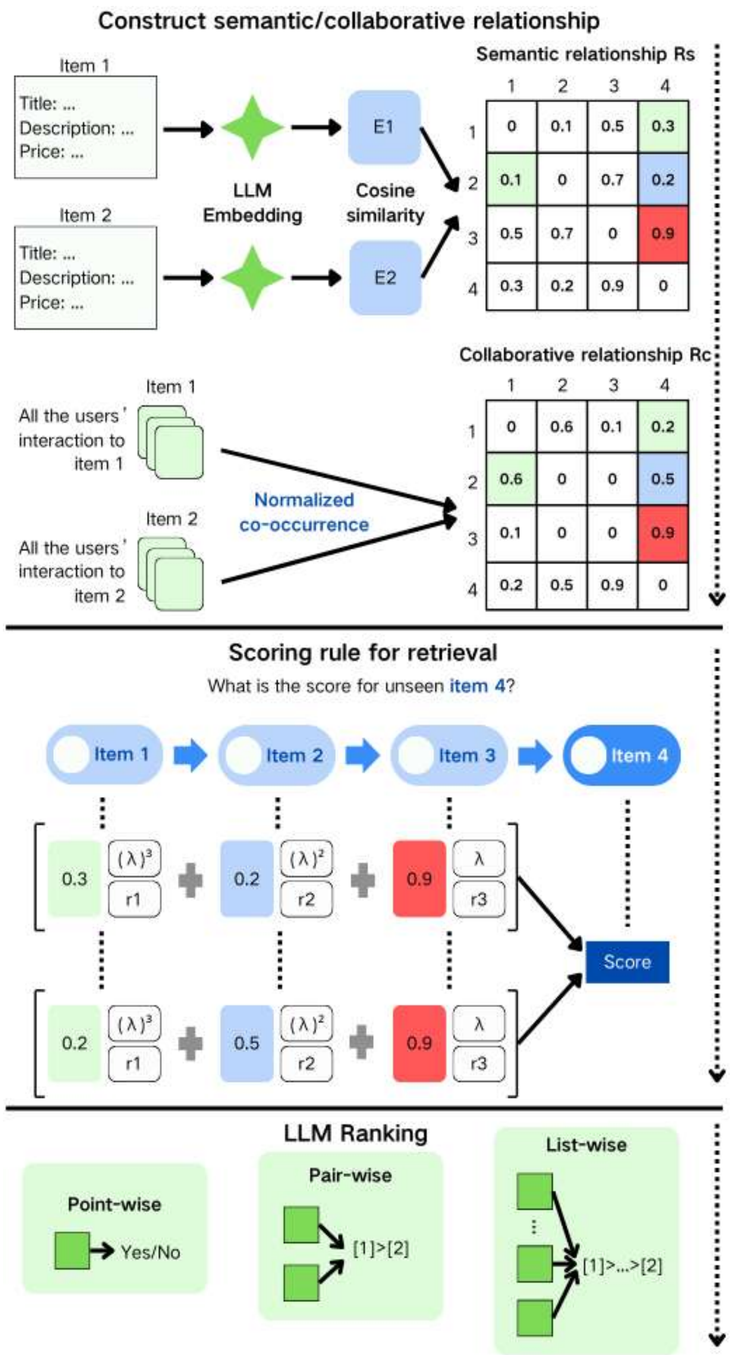
Motivation



SmartRec is a recommendation system framework that **does not require additional training**. It utilizes **the semantic embeddings** from large language models (LLMs) and **collaborative information from user behaviors** to generate **high-quality recommendations.**

Moreover, it achieves comparable or even **better results** than fully fine-tuned systems on multiple datasets.

Theoretical method



1. Semantic and Collaborative Relationship Construction

Semantic **Relationship:**

1. Construct prompts by inputting item information and metadata into the LLM embedding model, obtain de-dimensional embedding vectors of items, and form an embedding matrix E.
2. Calculate the semantic relationship between the embedding vectors Eia and Eib of two items (ia, ib) using cosine similarity to obtain a semantic relationship matrix Rs.

Collaboration relationship:

1. To understand collaborative relationships, consider how often users interact with item combinations. For each item i, derive a sparse user-interaction array, forming a set C of user-item interaction arrays.
2. Compute the collaborative relationship between two items using cosine similarity of their sparse arrays Cia and Cib, yielding a typically sparse collaborative matrix Rc.

2. Retrieval Scoring Rule

Formula:

$$\text{score}(x) = \frac{1}{n}\sum_{j=1}^{n} r_j \lambda^{t_j}\left[aR_S^{xj} + (1-a)R_C^{xj}\right]$$

where $\alpha = 0.5$ balances semantic and collaborative weights.

- R*s^xj* and R*c^xj* denote the semantic and collaborative relationships between item x and user's historical item s*j*;
- r*j* is the user's rating for item s*j*;
- λ^*tj* is an exponential decay function for the time order *tj* of the item in the user's historical sequence(latest item: *tj*=1; oldest item: max *tj*=n);
- a is a weight factor.
- Finally, output the top k items in descending order of scores.

3. LLM Ranking Stage

Use the LLM to further rank the top-k retrieved items for better recommendation quality. Incorporate retrieval-stage order information to leverage the LLM's strengths in understanding user preferences, capturing item relationships, and contextual relevance.

1) Point-to-point sorting: The LLM outputs the correlation (yes/no) or score (for example, Item 1 gets 0.8 and Item 2 gets 0.6) for a single item.

2) Pairing sorting: By sliding the window to compare adjacent items (such as Item 1 vs Item 2), the LLM determines the priority (such as [1] > [2]).

3) List sorting: Globally sort multiple items (such as [1,2,3]), and LLM optimizes the order through A sliding window (such as [1] > [2] > [3]).

Experiment & Results

Data statistics

Dataset	#Users	#Items	#Interactions	#Density
Beauty	965	53	5189	0.1196
Software	3241	4457	24650	0.0052
Prime_Pantry	15388	7841	150402	0.0231

Annotation:Density represents the percentage of non-zero elements in the uses_items matrices.

Dataset source:
The experimental data employed in this study are derived from the 2018 Amazon Reviews dataset.

Design Logic of Sub-datasets:
Sub-datasets are ordered by increasing size, a deliberate selection informed by preliminary observations that dataset scale exerts a significant influence on runtime performance metrics. This design enables systematic comparison across varying data volumes, facilitating exploration of scale-related effects.

Filtering approach:
Users and items with fewer than five interaction records are filtered out.To ensure comparability with existing literature, the data processing pipeline adheres to established protocols in prior research.

Result Comparison

KNN Baseline Method

Metric	All Beauty	Software	Prime Pantry
hit@10	0.0895	0.0039	0.0004
hit@5	0.0593	0.0017	0.0001
ndcg@10	0.051	0.0015	0.0002
ndcg@5	0.0413	0.0007	0.0001

Retrieval

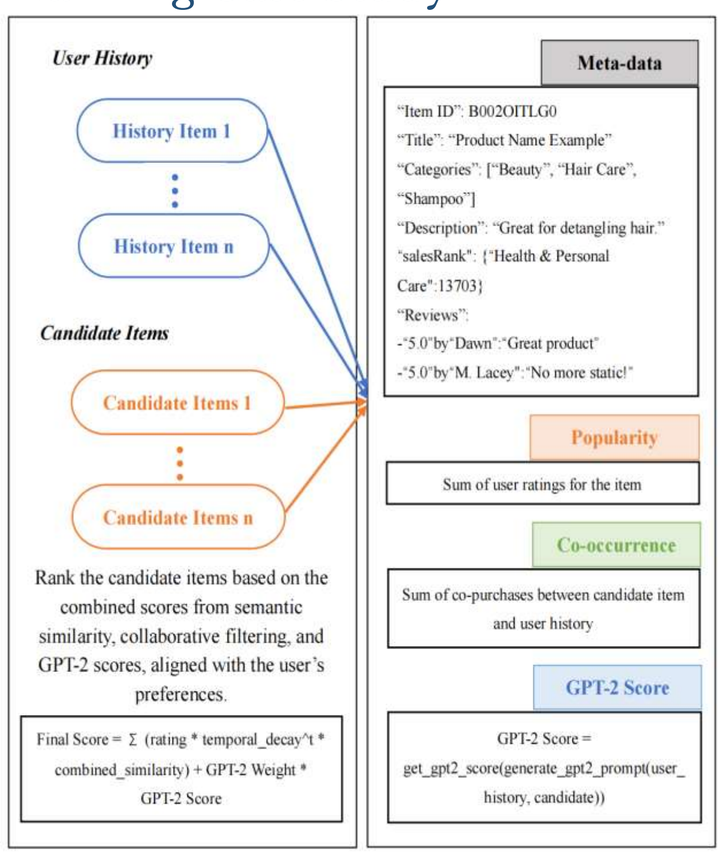
Metric	All Beauty	Software	Prime Pantry
hit@10	0.4788	0.1166	0.0508
hit@5	0.4383	0.0836	0.0358
ndcg@10	0.2260	0.0683	0.0297
ndcg@5	0.2126	0.0577	0.0249

Retrieval and Score

Metric	All Beauty	Software	Prime Pantry
hit@10	0.4788	0.1166	0.0510
hit@5	0.4456	0.0568	0.0276
ndcg@10	0.2522	0.0527	0.0296
ndcg@5	0.2413	0.0332	0.0172

Compared to the traditional KNN baseline, the new STAR model **achieves a massive leap in performance**, but the effectiveness of its "Score" refinement module is highly dependent on specific dataset characteristics and is not universally applicable.

Ranking Case Study



The system ranks **candidate items** by integrating **user history**, **item metadata**, **popularity**, **co-occurrence**, and **GPT-2 semantic scores**. ***For example***, if the user bought **shampoo**, the model scores across **dimensions** to recommend the most **relevant haircare product**.

Reference List & Github Program

Lee, D.-H., Kraft, A., Jin, L., Mehta, N., Xu, T., Chi, E. H., & Yi, X. (2025). STAR: A Simple Training-free Approach for Recommendations using Large Language Models. ***arXiv preprint arXiv:2410.16458v2.***

This project focuses on the innovative research of recommendation systems, proposes an enhanced STAR recommendation system, and combines GPT-2 ranking technology. The core code and detailed content can be viewed on https://github.com/Lexie-pq/STAR-Rec-GPT2.